

Preserving Source Location Privacy against the Global Attacker Hiding in FOG

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Abstract—FOG computing provides real-time service for analyzing data sensed by massive wireless embedded devices. However, an attacker can get a global view of an event-driven sensor network through the contextual information collected in FOG. We present an attack model that uses such information to analyze traffic and infer the source location. To resist traffic analysis, dummy message injection approaches have been widely applied. In previous studies, a method in which all nodes in a global scope send packets periodically gains perfect privacy but fails to provide energy-efficient and low-latency service. We propose a novel scheme, DLSA (Dark-Light Stripe Alternation). Compared with state-of-the-art techniques, our methods can not only maintain perfect privacy but also reduce the communication overhead by more than 50% and the end-to-end latency by more than 40%.

Keywords—Wireless Sensor Network; global attacker; contextual information; location privacy; random walk

I. INTRODUCTION

FOG (From cOre to edGe) computing was delivered as a version by Cisco in 2012 [1]. Compared with cloud computing, this new distributed computing is closer to the ground, or rather close to large amounts of sensed data. FOG services are used by applications that require low latency and short-distance communications, which cannot be satisfied by cloud computing. In addition, one of the significant distinctions from cloud computing is that FOG services have mobility support. Therefore, FOG services can be widely adopted to manage or analyze data [2]. However, an attacker can take advantage of FOG to eavesdrop on a wireless network and illegally gain assets.

In some WSN applications, the location privacy of certain events must remain undetected even by analyzing the network traffic when the events are transmitted through the WSN to the monitor center (*sink*). A sensor can be designed as tiny as a particle of dust, which can be deployed in a very subtle way to avoid being visually positioned by the attacker, but an attacker can still predict the entire network topology according to sensors' transmission attributes. Moreover, attackers can be classified as local or global according to their attacking capability. Unlike a *local attacker* [4-5] who can only eavesdrop within a certain range and back-trace the routing path until it reaches the source, a *global attacker* [6-7], who learns the entire network traffic information, can make use of temporal consistency and the correlation between packets, such as the received signal strength indication (RSSI) and the

direction of the packets, the inter-packets time, the packet occurrence time, etc., to infer the source location.

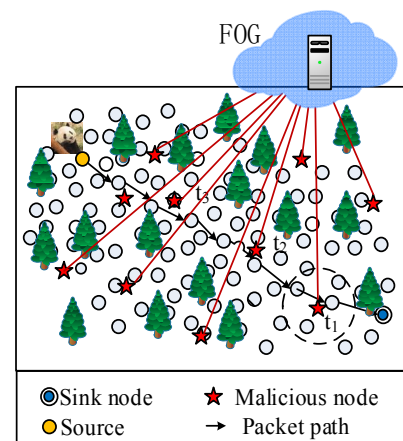


Fig. 1. A Global Attacker Hiding in FOG

To analyze the entire network traffic, a global attacker might not be a specific object, which can combine networking resources (including many distributed observers) and storage and computers). On carrying the signal detection equipment, each observer (including sensors and communicating equipment) can detect the strength and direction of the transmitted signals and can send the contextual information to a global attacker. For instance, in a WSN for monitoring endangered wild animals, as shown in Fig. 1, an electric sensor carried by a panda can send its own information to the sensors nearby and then be delivered to the sink. The first sensor that receives the signal of the monitoring asset, such as the panda, is called the *source node*. Many mobile observers (malicious nodes) are distributed in the forest, eavesdropping on the WSN. With the traffic analysis techniques collected by these observers, the global attacker will infer the network topology and construct the process of event transmission according to the contextual information. Therefore, the location privacy of the panda (source) is under substantial threat, and we must protect its location privacy during data transmission, thereby avoiding economic or resource loss.

As the attacker is invisible to the entire network, existing solutions can enhance privacy at the cost of a significant increase in energy consumption and delay. For a global attacker with topological routing information, it is easier to analyze the network traffic [8]. Although the STaR method [9]

uses the intermediate node as bait, a global adversary can still easily pinpoint the source and sink, and its low performance in privacy will be shown by comparison in our experimental section. When the privacy methods do not use dummy packets, such as the phantom routing, the global attacker can obviously pinpoint the location of the source node, based on the earliest transmission time of a phantom flooding. Even a local attacker with a hearing range more than three times that of the sensors, the capture likelihood is as high as 97%. Clearly, if all the traffic in the network is real event messages, it is unlikely to achieve source privacy under such a strong attack model. Therefore, we use network-wide dummy messages to achieve global privacy. The most intuitive method for perfect privacy protection is to make each node send packets at a constant rate, regardless of whether the packet is a dummy or real [7]. The attendant problem is that it is infeasible to send packets at a constant rate in real applications, and the high rate causes a high-energy cost, whereas a low rate leads to heavy latency.

Polling solutions, such as the minimum connected dominating sets (MCDS) method [10], seem to achieve perfect privacy by confusing the adversary, with less energy cost after the selection of sensor nodes for the polling sets. Unfortunately, since the computation of a single MCDS is an NP-complete problem [11], once the network topology changes, reconstruction and reorganization will lead to a costly update. Moreover, if no active sensors are nearby when the event happens, it will cause serious delay of packets to the sink. Therefore, network connectivity is also an important topic in WSN [2, 10]. Generally, the network is assumed to be dense when the security routing is applied, namely, the relay node is always on the path. However, as wildlife reserves cover a large area, dense deployment of sensors would cause excessive redundancy and an increased budget, while sparse deployment leads to more serious problems such as terrible link quality.

To solve the above problems, we further study the approaches for hiding the contextual information of packets to oppose the analysis of source location privacy in an event-driven sensor network. In this article, we present a network partition model that makes the network more flexible. And we employ a global adversarial model, similar to the one assumed in [7, 10, 12]. Furthermore, we design the DLSA (Dark-Light Stripe Alternation) scheme for monitoring static objectives to improve the flexibility of the network based on the random walk mathematical model [15].

II. PROBLEM DISCRIPTION

A. Network Model

In a homogeneous wireless sensor network, there are few neighbors in a sparse network, and it is easier for an attacker to locate the sender nearest him, so the nodes of the network in our study are densely connected, which means that the attacker cannot visually infer the topology geographically by locating the presence of sensors with a global camera view.

From the view of the external attacker, the format and size of packet $\hat{P}_i$, where $i \in ID$, are the same. In addition, the content is encrypted based on an elliptic curve, which can only be decrypted by the sink. Each sensor node's identity

information is encrypted to prevent an attacker from revealing the content of the packets, therefore he cannot distinguish between true packets and pseudo ones. The packets are associated with a hash digest $H(\hat{P}_i)$, for every receiver to filter the dummy packets.

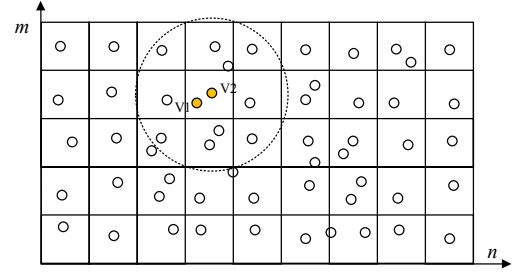


Fig. 2. Grid distribution

In our study, the network is further divided into cells by m rows and n columns, with at least a sensor node in each grid. There are N nodes $\{v_i | 1 \leq i \leq N\}$ and each sensor has the same computing and storage capacity, with each of the v_i nodes knowing its own location and the sink. Each node is labeled $N(ID, (c, r))$, where (c, r) indicates that node N_{ID} is in the r -th row and c -th column of the grid. For instance, $V1$ and $V2$ are all labeled in $N(V1, (c, r))$, and $N(V2, (c, r))$, respectively, and they are in the same cell or grid (c, r) , as shown in Fig. 2. Here, c and r are integers, where $0 < c \leq n$, $0 < r \leq m$, and m, n are the number of cells to measure the length and width of entire network. However, the size of each cell is fixed and preset according to the practical application, which must require the condition that a node in a grid can communicate with all the neighbors in the nearest 8 grids within the communication range, as shown in Fig. 2. In addition, each node needs at least one neighbors in each direction. In the same grid, there may be multiple nodes, so every node is identified by its own ID. In fact, the coordinate direction is randomly selected. The period of the network is Δ , namely, at the end of each Δ , no more than one packet is sent. All data will be encrypted, and it is assumed that an attacker cannot know the data.

B. Attack Model

The attacker in this article is (1) A **passive and external** listener who will not have any substantial effect on the function of the network; (2) A **well-equipped** attacker with instruments such as a spectrum analyzer, which, according to the transmission angle and signal strength of the received data, determines the location of the sender, and then moves directly to it; (3) A **global** attacker who cannot visually find the sensors but knows all the traffic in the network. and (4) An attacker with **infinite computing and power resources**, like a command center, who can deduce the source location by analyzing the global network traffic information observed and uploaded by each eavesdropper through the entire WSN.

Specifically, according to the packet rate of the global network, and the spatial and temporal correlation between packets, the attacker can infer the location of the data source and the destination node, revealing the location of the monitoring asset. For each observer $\alpha_i \in A$, he capture the

information $O_{ai}(H(\dot{P}_i), t(\dot{P}_i), \xi(a_i))$, when he receives the packet $\dot{P}_i$ at time t , where $\xi(a_i)$ is the area that an attacker observes. Generally, once the signal is intercepted, the attacker can determine the approximate location $\hat{i}(V_i)$ of the sender and the grid location $V_i(r, c)$ where the sender belongs, according to the receiving antenna angle and the strength of the signal. Once the location of the target of interest is confirmed, the attacker moves to the location to check whether it is true. If it is, the observer will capture the target successfully, otherwise he will move to a new location for detection by real-time interactions supported by FOG application. Before an attacker performs traffic analysis, it is necessary to pre-process the data provided by different observers.

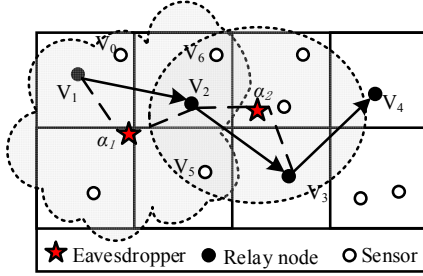


Fig. 3. Heterogeneous Eavesdroppers intercepting packets globally

In the analyzing process, there are three basic steps. The first step is elimination of redundancies. In time t ($t \ll T$), there are some repeated data intercepted by observers. As shown in Fig. 3, both α_1 and α_2 intercept the packets from V_2 , $O_{a1}(H(\dot{P}_i), t(\dot{P}_i), \xi(a_1))$ and $O_{a2}(H(\dot{P}_i), t(\dot{P}_i), \xi(a_2))$, respectively. According to the observer's eavesdropping range $\xi(a)$, the RSSI (received signal strength indicator) and the receiving direction, the attacker can infer the location of the packet: $\hat{i}(\dot{P}(\alpha_1)) = \hat{i}(\dot{P}(\alpha_2)) = V_2(c, r)$, and $t(\dot{P}(\alpha_1)) = t(\dot{P}(\alpha_2))$. Thus, $\dot{P}(\alpha_1) = \dot{P}(\alpha_2)$, indicating that the observers α_1 and α_2 intercept the same packet from V_2 , and the redundant packets are removed. In this scenario, α_1 can distinguish V_2 from V_3 , because V_2 and V_3 are not within a cell, but α_2 can not distinguish V_2 from V_6 , for $\hat{i}(V_2) = \hat{i}(V_6)$.

Second, we need to filter events. These non-repeated packets $O_{ai}(H(\dot{P}_i), t(\dot{P}_i), \hat{i}(V_i), \hat{E}_k)$ have been sorted by the intercept time t , after the first step. For example, in Fig. 3, the adversary has obtained the estimated location of V_1, V_2 and V_3 . Because the location V_2 and V_6 can not be distinguished, namely, $\hat{i}(V_2) = \hat{i}(V_6)$, the distance from V_1 to V_2 is the value between $D_{max} = \max(d(V_1, V_2), d(V_1, V_6))$ and $D_{min} = \min(d(V_1, V_2), d(V_1, V_6))$. If the events P_1 and P_2 of observation sets captured from V_1 and V_2 respectively are the same, they must meet the condition that the difference of the capture time or the intercept time is $D_{max} < [t(\dot{P}_2) - t(\dot{P}_1)] \cdot \gamma < D_{max}$. Therefore, the attacker infers that the associated sets belong to the same event $\hat{E}_k$, labeled as: $O_{ai}(H(\dot{P}_i), t(\dot{P}_i), \hat{i}(V_i), \hat{E}_k)$.

$$T_{min} < t(\dot{P}_{i+1}) - t(\dot{P}_i) < T_{max} \quad (1)$$

Thresholds $T_{min}(D_{min})$ and $T_{max}(D_{max})$ reflect bounds on the minimum and maximum delays for relaying packets from $\hat{i}(u)$ to $\hat{i}(v)$:

$$T_{min}(D_{min}) = \text{delay} \left\lfloor \frac{d_{min}(\hat{i}(u) - \hat{i}(v))}{\gamma} \right\rfloor \quad (2)$$

$$T_{max}(D_{max}) = \text{delay} \left\lceil \frac{d_{max}(\hat{i}(u) - \hat{i}(v))}{\gamma} \right\rceil \quad (3)$$

where γ is the sensor communication radius. T is an estimate of the packet transmission delay between two hops. The lower threshold prevents false association of labels due to event concurrence. The upper bound associates transmissions that occur close in time and in space with the same event.

Finally, a topology approximation is obtained by establishing links between labels, so the adversary can infer the source and sink location, the routing path to the sink, and the occurrence times associated with the same event. For the same event, packets are arranged in chronological order, according to the time correlation. The sender where the packet is captured in the minimum amount of time is usually the source node, and as multiple events are delivered to the sink, the source location will be repeatedly confirmed and verified.

C. Privacy Model

For a location privacy preserving scheme, the degree of privacy can be achieved in terms of information entropy to measure[7], and the criterion is defined as follows:

Definition1: (Privacy Criterion): Let $\Omega \in R^n$ be the privacy information of interest, namely, the privacy assets such as pandas, and $\tilde{O}$ be the information deduced by the attacker, assuming that the nodes in $\tilde{O}$ are equally likely to be the privacy objects. $\tilde{O} \in R^n$. The probability of the source node in $\tilde{O}$ can be estimated as $P(\tilde{O}) * |\Omega| / |\tilde{O}|$, where $P(\tilde{O})$ is the probability of capturing source node in the area of $\tilde{O}$. Here, we formally define the privacy criterion:

$$\Theta = \frac{1}{P(\tilde{O})} \sum_{|\tilde{O}|} \left(-\frac{1}{|\tilde{O}|} \log_2 \frac{|\Omega|}{|\tilde{O}|} \right) = \frac{1}{P(\tilde{O})} \cdot \log_2 \frac{|\tilde{O}|}{|\Omega|} \quad (4)$$

As Θ increase, the privacy of network is better. Intuitively, the denser the sensor distribution, the higher the degree of privacy. Assuming that the probability of events occurring near a node is the same, privacy is inversely proportional to the probability, namely, if $P(\tilde{O})$ is small, the privacy of each cell is high. Furthermore, the higher the value of the privacy criterion is, the better privacy the network has.

As for the same number of nodes with different spatial distributions covering the same area of a square space, the nodes in Fig.4(a) are deployed randomly in the field, while Fig.4(b) shows the grid distribution of nodes, where the distance between the nodes is equal. Fig. 4 shows two sensor nodes that are captured as the source node in a squared field, with one node's sensor area in blue with a size of $S_{(blue)}$, and the remaining area is red, with a size $S_{(red)}$. If there is only one source in the squared area, thus the probability of source node captured is:

$$P(\vec{O}) = \frac{1}{2S_{(blue)}} + \frac{1}{2S_{(red)}} = \frac{S_{(blue)} + S_{(red)}}{2S_{(blue)} \cdot S_{(red)}} \quad (5)$$

$P(\vec{O})$ of the uniform distribution in Fig.4 (b) is greater than that of the random distribution in Fig.4 (a). If $S_{(blue)} + S_{(red)}$ is the given squared area S , $S_{(blue)} \cdot S_{(red)} = S_{(blue)} \cdot (S - S_{(blue)})$, only when $S_{(blue)}$ approaches to $S/2$, the value of $S_{(blue)} \cdot S_{(red)}$ is maximum. Therefore, in a fixed space, with a given number of sensor nodes, $P(\vec{O})$ is a minimum of the grid distribution when the node distribution is uniform.

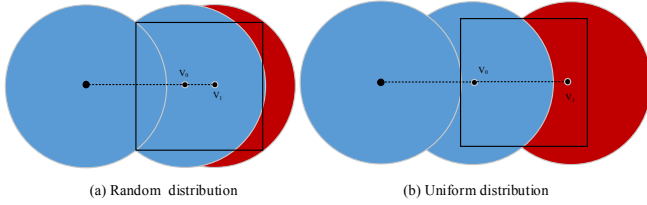


Fig. 4. Node distribution

III. OUR SOLUTION

A. Dark-Light Stripe Alternation

In this section, we propose a method for resisting the traffic analysis of the global attacker for source location privacy. The method proposed in this paper is applied for a static object. At the end of the scheme, we analyze the privacy and network overhead.

In an event-driven WSN, the source should report the abnormal situation to the control center in real-time. Normally a shortest-path routing [4,5,8] is applied due to the shortest delay and the least energy consumption. It is reasonable to assume that with dense deployments in which the shortest paths can be approximated by straight line. Assuming the object monitored is static, we arrange the source node and the sink node in the same column and divide the remaining network in parallel with the column based on the grid network model presented in Section 3.1. The nodes that belong to the adjacent columns are classified into different sets. For an arbitrary node $N(ID, (c, r))$, here we have sets $\mathcal{I}$ and $\tilde{\mathcal{I}}$:

$$\mathcal{I} = \{N \mid 0 \equiv c \pmod{2}, 0 < c \leq n\} \quad (6)$$

$$\tilde{\mathcal{I}} = \{N \mid 0 \equiv (c+1) \pmod{2}, 0 < c \leq n\} \quad (7)$$

Each sensor node knows the location of the sink ($S, (c_s, r_s)$) and its own grid, but only the source node knows the coordinates of the source.

All silent the nodes fail to continue forwarding packets, even if they receive these packets and then discard them. All the silent nodes are labeled as set **dark**, while all the active nodes are labeled as set **light**, as shown in Fig. 5. Let an epoch $T = \mu\Delta$, where μ is an integer greater than 1. We partition the WSN to $\mathcal{I}$ and $\tilde{\mathcal{I}}$ that remain active in an alternating fashion. Let k be an integer and $k \in [0, +\infty]$, in the interval $(2kT, (2k+1)T)$, let the nodes in set $\mathcal{I}$ remain the **active** state, marked as light, and the nodes in set $\tilde{\mathcal{I}}$ remain the **silent** state. In the

time interval $((2k+1)T, (2k+2)T)$, let $\mathcal{I}$ be **dark** and $\tilde{\mathcal{I}}$ be **light**. To hide the location of the real sender, all the sensor nodes in the light state send a packet periodically at the end of each Δ . If no real packet is in memory, then a dummy packet is sent, and the dummy packet will be discarded by the receiver. The same traffic pattern can confuse the attacker, thus preventing him from distinguishing where an event has occurred. The ID of a node can only be decrypted by the sink, and the node is denoted $N(c, r)$.

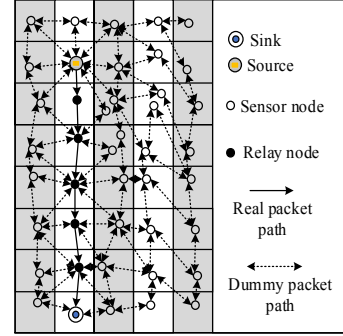


Fig. 5. Dark-Light Stripe Alternation

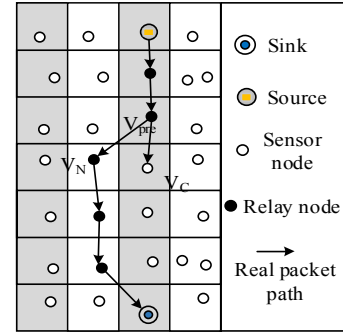


Fig. 6. Dark-Light Stripe Alternation

Real packet *RealMsg* is sent according to the delivery rules as follows:

Rule 1. No matter how many nodes are in a grid, only one is randomly selected as a relay node for each delivery, and it does not have to be the same one every time;

Rule 2. Regardless of whether the state of the current node $N_C(c, r)$ is light or dark, it can receive and store the real packet *RealMsg* broadcast from the neighbor $N_E(c_e, r_e)$. If a packet *RealMsg* is new, and $|r - r_s| < |r_e - r_s|$, then it will be stored by the current node, otherwise the packet will be discarded;

Rule 3. The range of packet delivery does not go beyond the adjacent eight cells.

Rule 4. If the current node N_C is in the light state, with the knowledge of $S(c_s, r_s)$, the data will be forwarded to the next-hop node $N_N(c_n, r_n)$, and deleted from memory after delivery. When $(r - r_s) < -1$, the source is the sink, and the next-hop node is $N_N(c, r-1)$. When $(r - r_s) > 1$, the next hop node is $N_N(c, r+1)$. When the $|c - c_s| = 1$, the algorithm will no longer choose the next-hop node, the current node directly sends *RealMsg* to the sink, and nodes automatically discard it.

Rule 5. If the current node $N_C(c, r)$ is in the dark state, and there is *RealMsg* in memory, and if $c < n/2$, then the algorithm sends the beacon signal to inform the neighbor node $N_{right}(c+1, r)$, otherwise to inform $N_{left}(c-1, r)$ of sending *RealMsg* in the memory to the sink.

Rule 6. If the source node is dark and has an event to send, it should wait until the next light epoch.

To ensure a minimum delay of the packet, the real packet will be delivered through the shortest route. The sink and the source node are arranged in the same column, and when this column is in the light epoch, the length of the shortest path h is the difference between the two ordinates, namely, $h = |r_s - r|$. Using the flow coordination method mentioned below, if the remaining time in the light epoch is greater than Δ , then the transmission to the sink can be completed in time. When the state is switching to dark during the process of transmission, all the nodes in the column where the source node and the sink are located are silent, but the nodes in the adjacent columns are active, and the sink can receive packets at any epoch, so packets are transmitted by the nodes in the adjacent column to the sink in the new epoch.

As shown in Fig. 6, when looking for the next hop, if V_C finds that its direct neighbors on the shortest path become dark, then it will notify V_N through a beacon signal to deliver the real packet. Because V_N has saved the packet from V_{pre} , it continues to deliver the real packet to the sink according to the rules above. To ensure that the source is delivered correctly even on the boundary, referring to Rule 5, the network is divided into two parts. If the source is on the left, the right column is selected to transfer data, and vice versa.

B. Latency Analysis

At the end of each Δ every light sensor will send a packet, true or false. Once a real packet arrives, it will be sent in priority instead of dummy packets. Suppose that starting from the source node, a real packet is transmitted to the sink by h hops. The latency of delivery is related to the path length, the size of the package, the quality of wireless communications and Δ . A larger Δ leads to less communication, but more latency. However, Δ can be set appropriately according to the specific application. The transmission process of a packet is illustrated in Fig. 7.

In the worst case, the packet arrives at the next hop when a packet is being delivered. Therefore, the packet must wait until the next Δ , and because the path length is h , the latency after h hops is $h \cdot \Delta$. If the epoch changes during the transmission, another epoch $T = \mu \Delta$ is needed, and the worst latency is $h \cdot \Delta + T = \Delta \cdot (\mu + h)$ from the source to the sink. In the best case, the real packet is delivered immediately after it arrives at the next hop. The average latency will be discussed below.

In fact, the drift of the network synchronization will also lead to latency, which is beyond the scope of this article. For the snapshot shown in Fig. 8, it is assumed that the clock drift is less than Δ . This is equivalent to the case in which a packet is sent in an arbitrary time within each Δ , as shown Fig. 8, and each transmission is independent.

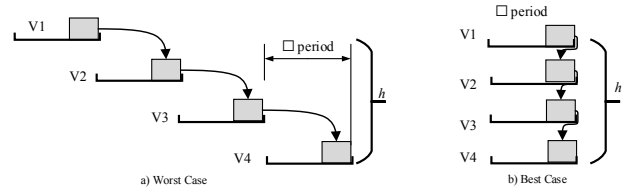


Fig. 7. The worst and best latency

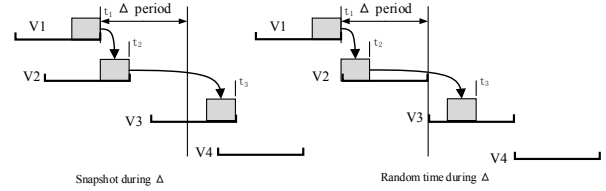


Fig. 8. Average latency

Proposition 1: when one packet is sent in each interval Δ , the average delay is: $0.58 \cdot h \cdot \Delta$.

Proof: As shown in the snapshots of Fig. 8, a packet, starting from V_1 , goes through B hops, and the delivery time of each hop is $t_1, t_2, \dots, t_B$. After Δ , the average number of hops that a packet travels from V_1 is $\bar{U}$. The value of $\bar{U}$ depends on the sequence of $t_1, t_2, \dots, t_B$, under $B!$ combination with equal probability, this is because the clock offset of each node within each Δ is random and independent. Thus, for $t_1 < t_2 < \dots < t_B$, the probability of random variable that a packet traverses in B hops during Δ is as follow:

$$P(\bar{U} = b) = \frac{1}{b!} \quad (8)$$

In the same way, for $t_1 < t_2 < \dots < t_{B-1}$, the probability of random variable that a packet traverses in $B-1$ hops during Δ is as follows:

$$P(\bar{U} = b - 1) = \frac{1}{b!} \left(\frac{b!}{(b-1)!} - 1 \right) \quad (9)$$

For an arbitrary number of hops $\beta \leq B - 1$, we obtain,

$$P(\bar{U} = \beta) = \frac{1}{b!} \left(\frac{b!}{\beta!} - \frac{b!}{(\beta+1)!} \right) = \frac{\beta}{(\beta+1)!} \quad (10)$$

Thus, the expectation of $\bar{U}$ is computed:

$$E(\bar{U}) = \frac{b}{b!} + \sum_{\beta=1}^{b-1} \frac{\beta^2}{(\beta+1)!} = \frac{1}{(b-1)!} + \sum_{\beta=1}^{b-1} \frac{\beta^2}{(\beta+1)!} \quad (11)$$

By applying the ratio test of series on the second term of $E(\bar{U})$, we obtain

$$\lim_{\beta \rightarrow \infty} \frac{\frac{(\beta+1)^2}{(\beta+2)!}}{\frac{\beta^2}{(\beta+1)!}} = \lim_{\beta \rightarrow \infty} \frac{(\beta+1)^2 (\beta+1)!}{\beta^2 (\beta+2)!} = \lim_{\beta \rightarrow \infty} \frac{\beta^2 + 2\beta + 1}{\beta^3 + 2\beta^2} \leq 1, \quad (12)$$

Which proves $E(\bar{U})$ is a converging series. Finally, for all $B \geq 1$, the convergence value of $E(\bar{U})$ computed via numerical analysis is 1.7183. In conclusion, in the time Δ , the average number of hops that a packet traverses from V_i is 1.7183. Thus, after h hops the average latency is: $\tau = \frac{h \cdot \Delta}{E(\bar{U})} = 0.58 \cdot h \cdot \Delta$

Ideally, the minimum latency for the shortest path is Δ , whereas the latency is apparently related to the length of the shortest path h . When the downstream packet is transmitted before a real packet arrives, the worst latency is $h \cdot \Delta + T = \Delta(\mu + h)$, and in the worst case that the source node needs to wait for the next epoch. By the proposition 1 proved above the average latency is $0.58 \cdot h \cdot \Delta$.

C. Privacy and energy

First, we analyze the level of privacy achieved using the DLSA (dark-light stripe alternation) scheme. According to Kerckhoffs' principle, the attacker knows exactly how the network is classified and the alternative scheme. Suppose that there is an event $\tilde{E}(\bar{P})$ occurring at time T , if $N_{source}(c_{source}, r_{source})$ is at a light epoch, it will send a real packet $\bar{P}$ instead of dummy packets at $t(\bar{P})$, where $t(\bar{P}) = T$, whereas if N_{source} is dark at time T , $\tilde{E}(\bar{P})$ will be sent in the next epoch at $t(\bar{P})$, where $t(\bar{P}) \neq T$. According to the attack model in Algorithm 1, the attacker cannot infer which set the source node belongs to. It is impossible to determine in which epoch the source node sends $\bar{P}$. Therefore, because the global attacker cannot classify $\bar{P}$ into a valid event $\tilde{E}$, the location of $\tilde{E}(\bar{P})$ at $t(\bar{P})$ is protected.

Because the packet is encrypted, true and false data packets are the same from the attacker's perspective. All the nodes in the light status will send packets, so the attacker O_a cannot detect the location of real packet $i(\bar{P}(O_{ai}))$ from the captured packets $O_{ai}(H(\bar{P}_i), t(\bar{P}_i), \xi(\alpha_i))$. Furthermore, the attacker cannot identify the exact source instead of a grid, as mentioned in Section 3.3. If the source location is protected, the corresponding sink location will also be protected.

IV. CONCLUSION

In this study, we propose a novel schemes DLSA for resisting a global attacker based on a grid partitioning of the network. DLSA can not only maintain perfect privacy, but also reduce the communication overhead by more than 50%, and the end-to-end latency by more than 40%. It makes use of polling different sets of nodes to reduce the network overhead significantly while ensuring the source location privacy. Our method is targeted for a dense connected network. However, in practice, there may be more complicated scenarios, such as holes occurring in the network. Therefore, we will study the robustness of the network in the future.

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REFERENCES

- [1] F. Bonomi, R. Mito, J. Zhu, and S. Addepalli, "Fog Computing and Its Role in the Internet of Things," in *Proceedings of the First Edition of the MCC Workshop on Mobile Cloud Computing*, New York, NY, USA, 2012, pp. 13–16.
- [2] K. Bhargava, S. Ivanov, C. Kulatunga, and W. Donnelly, "Fog-enabled WSN system for animal behavior analysis in precision dairy," in *2017 International Conference on Computing, Networking and Communications (ICNC)*, 2017, pp. 504–510.
- [3] D. Spiriyakin, A. Baranov, and V. Sleptsov, "Design of Smart Dust Sensor Node for Combustible Gas Leakage Monitoring," 2015, pp. 1279–1283.
- [4] P. Kamat, Y. Zhang, W. Trappe, and C. Ozturk, "Enhancing Source-Location Privacy in Sensor Network Routing," in *25th IEEE International Conference on Distributed Computing Systems, 2005. ICDCS 2005. Proceedings*, 2005, pp. 599–608.
- [5] Zhou Qian, Qin X L, Ding Y W, "Preserving source-location privacy efficiently based on attack-perceiving in wireless sensor network," *Journal on Communications*, vol. 39, no. 1, pp. 101–116, 2018.
周倩, 秦小麟, 丁有伟, "基于攻击感知的能量高效源位置隐私保护算法," *通信学报*, vol. 39, no. 1, pp. 101–116, 2018.
- [6] M. Chaudhari and S. Dharawath, "Toward a statistical framework for source anonymity in sensor network using quantitative measures," in *2015 International Conference on Innovations in Information, Embedded and Communication Systems (ICIECS)*, 2015, pp. 1–5.
- [7] K. Mehta, D. Liu, and M. Wright, "Protecting Location Privacy in Sensor Networks against a Global Eavesdropper," *IEEE Trans. Mob. Comput.*, vol. 11, no. 2, pp. 320–336, 2012.
- [8] B. Di Ying, D. Makrakis, and H. T. Mouftah, "Anti-traffic analysis attack for location privacy in WSNs," *EURASIP J. Wirel. Commun. Netw.*, vol. 2014, p. 131, Aug. 2014.
- [9] L. Lightfoot, Y. Li, and J. Ren, "Preserving Source-Location Privacy in Wireless Sensor Network Using STaR Routing," in *2010 IEEE Global Telecommunications Conference GLOBECOM 2010*, 2010, pp. 1–5.
- [10] A. Proano, L. Lazos, and M. Krunz, "Traffic Decorrelation Techniques for Countering a Global Eavesdropper in WSNs," *IEEE Trans. Mob. Comput.*, vol. PP, no. 99, pp. 1–1, 2016.
- [11] M. R. Garey and D. S. Johnson, *Computers and Intractability: A Guide to the Theory of NP-Completeness*. New York, NY, USA: W. H. Freeman & Co., 1979.
- [12] B. Alomair, A. Clark, J. Cuellar, and R. Poovendran, "Toward a Statistical Framework for Source Anonymity in Sensor Networks," *IEEE Trans. Mob. Comput.*, vol. 12, no. 2, pp. 248–260, 2013.
- [13] P. Erdős and P. Révész, "Problems and Results on Random Walks," in *Mathematical Statistics and Probability Theory*, P. Bauer, F. Konecny, and W. Wertz, Eds. Springer Netherlands, 1987, pp. 59–65.
- [14] William Feller, *An Introduction to Probability Theory and its Applications*, vol. 1. John Wiley & Sons Inc., 1957.
- [15] M. Shao, W. Hu, S. Zhu, G. Cao, S. Krishnamurth, and T. La Porta, "Cross-layer Enhanced Source Location Privacy in Sensor Networks," in *6th Annual IEEE Communications Society Conference on Sensor, Mesh and Ad Hoc Communications and Networks, 2009. SECON '09*, 2009, pp. 1–9.